

Lab 1: Descriptive Paper

Miguel Looney

CS 411

Thomas J. Kennedy

28 September 2026

Version 1

**Table of Contents**

|                                                 |   |
|-------------------------------------------------|---|
| 1 Introduction.....                             | 2 |
| 2 Gold Star Product Description .....           | 2 |
| 2.1 Key Product Features and Capabilities ..... | 2 |
| 2.2 Major Components (Hardware/Software).....   | 3 |
| 3 Identification of Case Study.....             | 4 |
| 4 Glossary .....                                | 6 |
| 5 References.....                               | 9 |

**Listing of Figures**

|                                                    |   |
|----------------------------------------------------|---|
| Figure 1: Major Functional Component Diagram ..... | 4 |
|----------------------------------------------------|---|

**Listing of Tables**

**No table of figures entries found.**

## **1 Introduction**

Attention-Deficit/Hyperactivity Disorder (ADHD) affects many adults, and can make it difficult to build routines, stay organized, and manage time (CDC, 2024). These challenges can make everyday responsibilities harder to complete and create the need for tools that help users stay on track. Gold Star was created to provide a simple and flexible solution to these problems.

Gold Star is a web-based habit-building and time management platform designed for individuals with ADHD and ADHD-like traits. It helps users organize tasks, build routines, and manage their schedules through an adaptive system that adjusts when tasks are missed or priorities change. This prototype demonstrates the application's main features and how they can support productivity and long-term routine building.

## **2 Gold Star Product Description**

Gold Star is a web-based productivity and habit-building platform designed primarily for individuals with ADHD struggling with routine creation and maintenance. Many people with ADHD struggle with organization, time management, and keeping up with daily responsibilities. Gold Star gives users a low-stakes environment where tasks, schedules, and habits are monitored without feeling overwhelmed. Gold Star is intended for easier routine development and maintenance, while offering support for users struggling to preserve long-term goal consistency.

### **2.1 Key Product Features and Capabilities**

Gold Star uniquely focuses on helping users organize tasks and schedules through an Adaptive Scheduling system which adjusts to missed tasks and changing priorities. Gold Star evolves with the user, reducing the frustration of falling behind, making it easier to stay on track. It utilizes techniques such as habit stacking, activity alternation, and the Pomodoro technique which are recommended for people that struggle with ADHD-like symptoms. The user can choose their productivity level and alter their schedule at any time throughout the day, giving them the

flexibility they need to feel successful. They can group tasks by importance level, allowing Gold Star to become personalized to their unique scheduling needs.

Gold Star rewards the user for completing tasks and milestones, giving them a sense of accomplishment. It does not punish them for missing goals, instead featuring a “Roll Over” system, allowing the user to move their missed tasks to another day. Combining flexibility and structure, the platform helps users overcome common ADHD challenges such as task initiation, consistency, commitment, and completion, ultimately improving productivity and reducing daily life stress.

## **2.2 Major Components (Hardware/Software)**

The system requires no specialized hardware. Users can access the application through any off-the-shelf device with a modern web browser. The server side runs on a single hosting environment supporting the Node.js web server and PostgreSQL database, with deployments being automatically pushed through GitHub. As shown in Figure 1, the system follows a three-tier structure: a Presentation Layer (the web-based UI), an Application Layer (the Node.js web server / API with authentication, and a notification handler), and a Data Layer (PostgreSQL database storing the encrypted user data, tasks, and routine data).

The software to be developed includes the UI /UX built with HTML, CSS, and JavaScript. It will feature a calendar view, routine dashboard, productivity level, and statistics dashboard. The Node.js web server handles routing, logic, authentication components, and the WebSocket-based notification handler. The PostgreSQL database stores user account information, tasks, routine data, and sensitive user information in an encrypted form. Development is supported by off-the-shelf tools: Visual Studio Code, Git and GitHub, GitHub Actions for CI/CD, Jest for testing, and Javadoc for documentation.

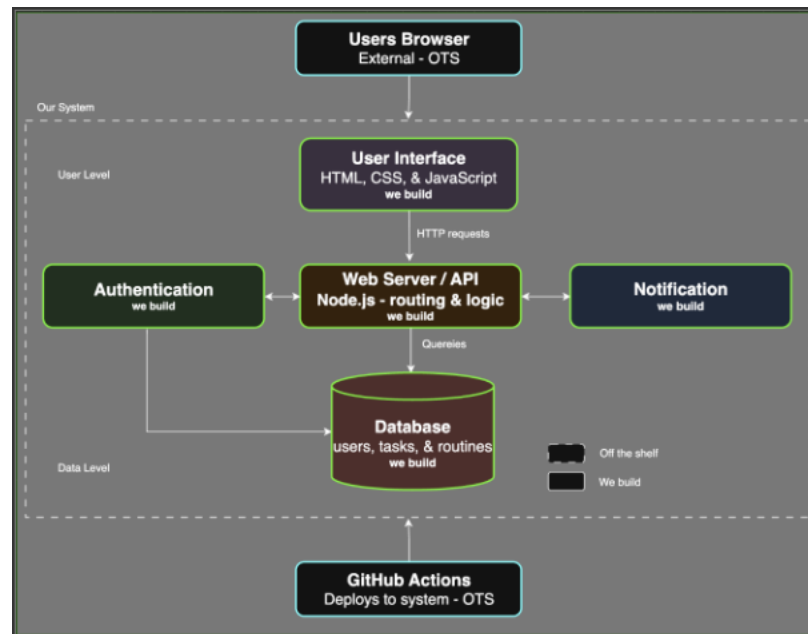


Figure 1: Major Functional Component Diagram

### 3 Identification of Case Study

Individuals with ADHD who struggle with time management, organization, and routine maintenance are Gold Star's primary intended demographic. Many people with ADHD face challenges during schedule creation and consistency due to difficulties with focus, motivation, and efficient task prioritization. To align with Gold Star's main purposes of automating task development, task maintenance, and schedule changes, this consumer demographic serves as the software's primary end user role intending to use the application for improvements in productivity and schedule balancing.

Despite individuals with ADHD serving as Gold Star's primary end user role, this software could also benefit other groups. Additional end user demographics may include parents, guardians, therapists, and other neurodivergent or neurotypical people seeking assistance with improving time management, productivity, and routine building skills. Gold Star was developed for individuals with ADHD, but the application is not intended to exclude users without either ADHD or ADHD-like traits. Regardless of symptoms, people who struggle with organization,

productivity, or habit-building consistency are expected to find this application useful for managing daily tasks and stable routines.

## 4 Glossary

- **Activity Alternation** – Going from an activity requiring high mental focus to an activity requiring low mental focus.
- **Activity Switching** – The cognitive process of disengaging from one task and shifting mental focus to another.
- **Adaptive Scheduling** – To schedule a task which can be completed anytime within a set timeframe. For example, a task which can be completed anywhere inclusively between 7:00 P.M. and 10:00 P.M.
- **Application Programming Interface (API)** - A set of standard rules, protocols, and definitions allowing different software applications to interact with one another.
- **Attention-Deficit/Hyperactivity Disorder (ADHD)** - A common neurodevelopmental disorder that alters the brain's ability to plan, focus, remember instructions, and manage emotions.
- **Cascading Style Sheets (CSS)** - A style sheet language used to determine the visual presentation, layout, and aesthetics of a website.
- **Git** - A digital version control system intended to preserve a history of saved modifications, additions, and deletions to software source code.
- **GitHub** - A cloud-based platform where developers can collaborate on software code.
- **Gold Star** - A web-based habit-building and time management platform designed specifically for individuals with ADHD.
- **Habit Stacking** – Interspersing new habits with existing habits to make task completion easier to complete without becoming overwhelmed.
- **Hyperactivity** - A state of being unusually or abnormally active.

- **HyperText markup Language (HTML)** - The standard markup language used to create and structure webpages.
- **Javascript (JS)** - A lightweight programming language used to build dynamic interactivity on web pages.
- **Jest** - An open-source Javascript testing framework.
- **JSDoc** - An API documentation generator and syntax standard for JavaScript.
- **Pomodoro Technique** – A time-management technique which separates work into 25-minute increments and 5-minute-long breaks. After repeating this technique 4 times, a longer break from 20 to 30 minutes is taken before the process repeats.
- **PostgreSQL** - An open-source RDBMS that stores, manages, and retrieves data.
- **Reinforce** - To make something stronger, more effective, or more compelling by adding support.
- **Relational Database Management System (RDBMS)** - Software used to store, manage, query, and retrieve data structured in table form.
- **Rigid Scheduling** - To schedule a high-priority task which must either be initiated at or completed before an established time. For example, a task which must start at 6:00 P.M. or must be completed by 6:00 P.M.
- **Routine Adaptation** – A requirement for an individual to change one's schedule or create new routines.
- **Routine** - A sequence of actions regularly followed, or a fixed program.
- **Task Deviation** – When a task has an unexpected change regarding completion. A task may be accomplished out of order in one's schedule, or it may not be accomplished at all.

- **Visual Studio Code (VSCode)** - A development environment used for software engineering and web development.

## 5 References

- ADHD Philadelphia. (2026, May 30). Why Adults With ADHD Struggle to Stay Consistent With Routines. ADHD Philadelphia. <https://www.adhdphiladelphia.com/blog/why-adults-with-adhd-struggle-to-stay-consistent-with-routines>
- Advanced Psychiatry Associates. (2026). Strategies for Adults Living With ADHD. Addicta. <https://advancedpsychiatryassociates.com/resources/blog/strategies-for-adults-living-with-adhd>
- CDC. (2024, May 29). ADHD Across the Lifetime. Attention-Deficit / Hyperactivity Disorder (ADHD). <https://www.cdc.gov/adhd/articles/adhd-across-the-lifetime.html>.
- Chen, J., Meng, Y., & Nie, K. (2026, March). “Not Just Me and My To-Do List”: Understanding Challenges of Task Management for Adults with ADHD and the Need for AI-Augmented Social Scaffolds. arXiv. <https://doi.org/10.48550/arXiv.2603.17258>
- Chhabildas, N. Pennington, B.F. & Willcutt, E.G. (2001, December). A Comparison of the Neuropsychological Profiles of the DSM-IV Subtypes of ADHD. *Journal of Abnormal Child Psychology*, 29, 529-540. <https://doi.org/10.1023/A:1012281226028>
- Kessler, R.C., Lane, M., Stang, P.E., & Van Brunt, D.L. (2008, April 21). The prevalence and workplace costs of adult attention deficit hyperactivity disorder in a large manufacturing firm. *Psychological Medicine*, 39(1), 137-147. <https://doi.org/10.1017/S0033291708003309>
- Kreider, C. M., Medina, S., & Slamka, M. R. (2019). Strategies for Coping with Time-Related and Productivity Challenges of Young People with Learning Disabilities and Attention-Deficit/Hyperactivity Disorder. *Children (Basel, Switzerland)*, 6(2), 28. <https://doi.org/10.3390/children6020028>
- Ogrodnik, M., Karsan, S., Malamis, B., Kwan, M., Fenesi, B., & Heisz, J. J. (2023). Exploring Barriers and Facilitators to Physical Activity in Adults with ADHD: A Qualitative Investigation. *Journal of developmental and physical disabilities*, 1–21. Advance online publication. <https://doi.org/10.1007/s10882-023-09908-6>
- Staley, B.S., Robinson, L.R., Claussen, A.H., Katz, S.M., Danielson, M.L., Summers, A.D., Farr, S.L., Blumberg, S.J., & Tinker, S.C. (2024). Attention-Deficit/Hyperactivity Disorder Diagnosis, Treatment, and Telehealth Use in Adults — National Center for Health Statistics Rapid Surveys System, United States, October–November 2023. *U.S. Centers for Disease Control and Prevention Morbidity and Mortality Weekly Report*, 70(40), 890-895. <https://www.cdc.gov/mmwr/volumes/73/wr/pdfs/mm7340a1-H.pdf>

